

Transformers In Computer Vision

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Agenda

- ViT - Attention for Visual Recognition
- DeiT - Compressing ViT via Distillation
- CLIP - Image to text embeddings
- Cross modal embeddings
- Auto encoder

Transformers In Computer Vision

Can we use Self-Attention in images ?



I: CNN Recap

The Architecture of CNNs

The Incumbent Champion

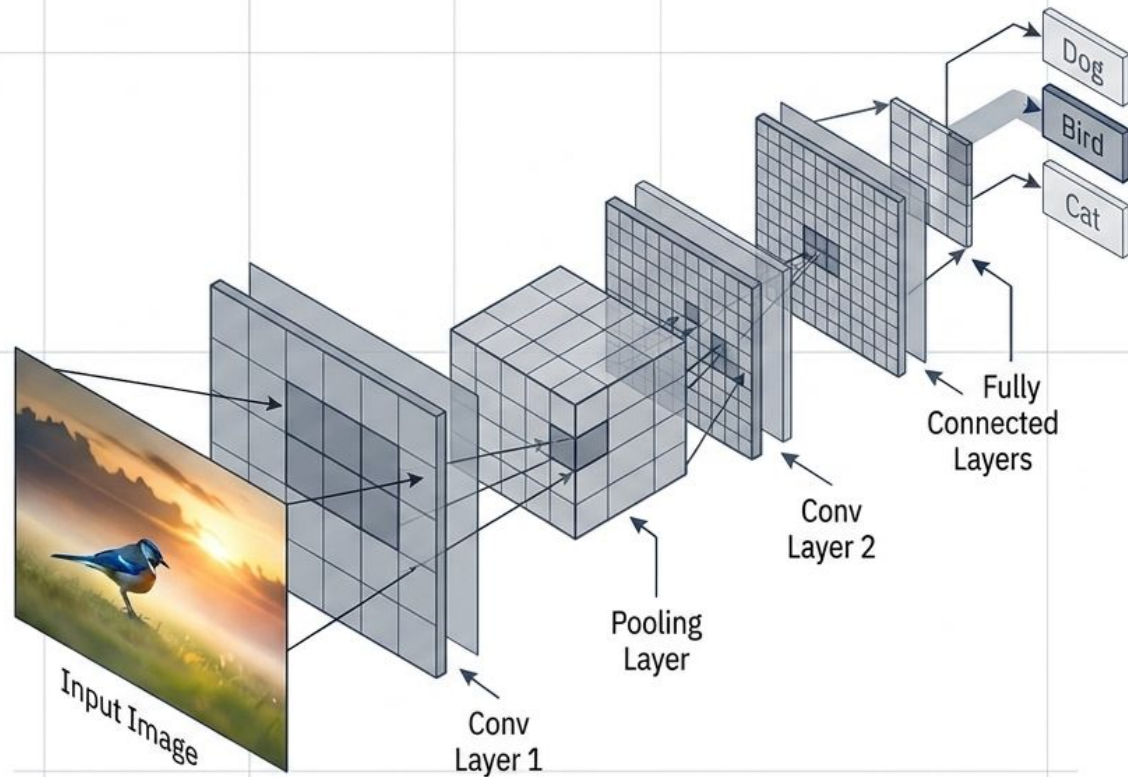
For years, Convolutional Neural Networks (CNNs) like ResNet and VGG ruled computer vision through an intuitive, elegant mechanism.

The Mechanism

Instead of viewing an entire image at once, CNNs utilize small computational filters that scan across the image grid.

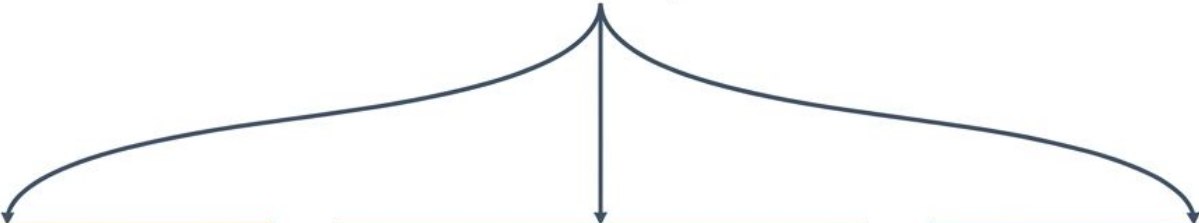
The Output

By sharing weights across space and stacking layers sequentially, they extract increasingly abstract features—from raw edges to complex objects.



The Three Pillars of CNN Architecture

CNNs have 3 key features



Pillar 1: Locality

- Assumption: Neighboring pixels are highly related; distant pixels are not.
- Function: Features are detected in small, heavily localized regions using convolutional kernels.

Pillar 2: Translation Invariance

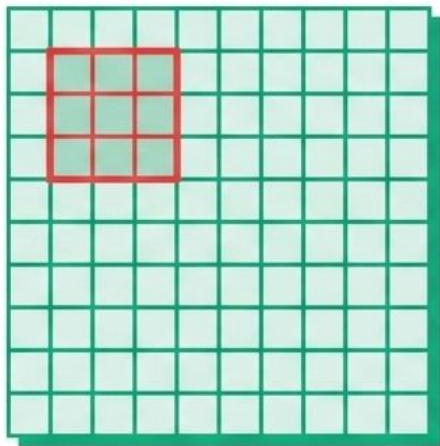
- Assumption: A feature's meaning is independent of its location.
- Function: A filter that detects a car in the top-left will detect the exact same car in the bottom-right.

Pillar 3: Weight Sharing

- Assumption: Visual patterns repeat across the canvas.
- Function: Filters are reused across all spatial locations to drastically reduce the total parameter count.

Q1: If Convolutional Neural Networks (CNNs) ruled computer vision for a decade, why do we need a new architecture?

A1: CNNs lack global context



Locality Limit: A 3x3 filter looks at only 9 pixels at a time, entirely missing the big picture.

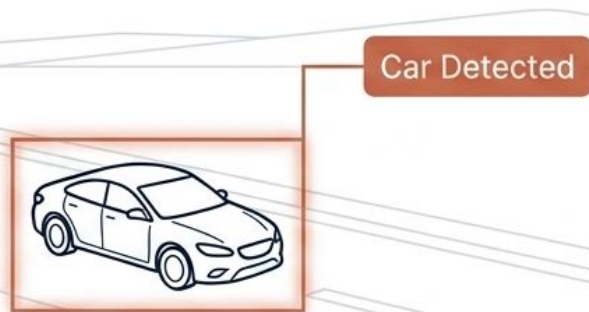


Translation Invariance Fail: CNNs detect localized 'car' features but miss the absurdity of the context. They assume an object is the same regardless of placement.

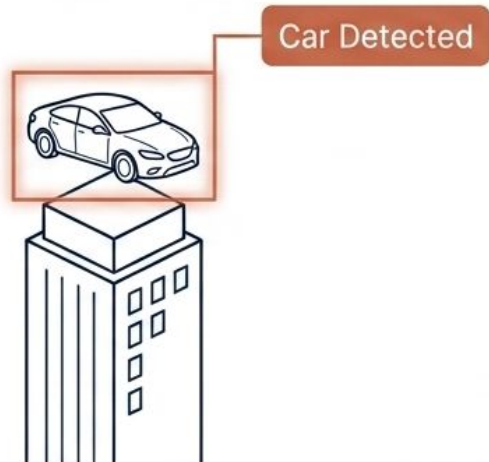
CNNs assume locality. To see the whole picture, they require stacking incredibly deep layers, leading to vanishing gradients. We needed an architecture that sees the entire image from layer one.

The CNN Dilemma: Context Blindness

Normal Context



Absurd Context



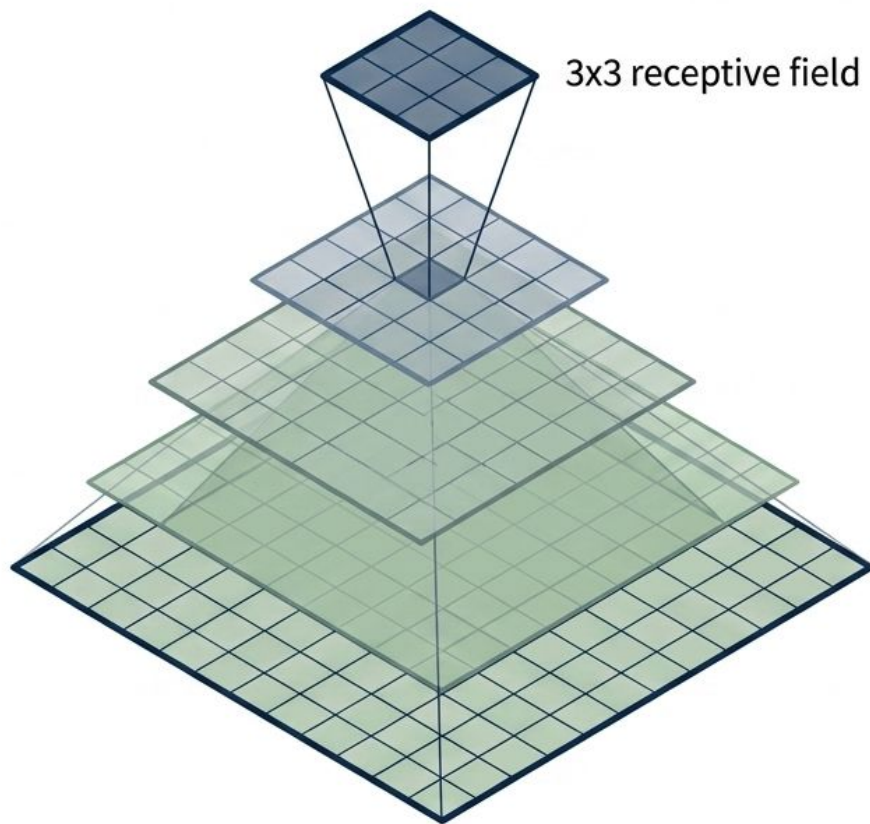
The Flaw in the Filter

Translation invariance guarantees a CNN will find a localized shape, but it fundamentally blinds the network to the surrounding reality.

Missing the Absurdity

Because the local "car-like edge" filter activates perfectly, the CNN confidently predicts "car." It lacks the global context to understand that cars do not belong on top of buildings.

The Receptive Field Bottleneck



The 3x3 Constraint

A standard CNN filter looks at a mere 9 pixels at a time.

The Scaling Problem

To understand an entire image, a CNN must stack hundreds of layers so a deep node's receptive field expands enough to encompass the whole picture.

The Mathematical Consequence

1. Exorbitant training times.
2. Degradation of the learning signal (Vanishing Gradients).

II: Vision Transformers

Vision Transformer (ViT): A new paradigm

The **Vision Transformer** (ViT), introduced by Google Brain in the paper “*An Image is Worth 16x16 Words*”, does exactly that.

Published as a conference paper at ICLR 2021

“AN IMAGE IS WORTH 16X16 WORDS: TRANSFORMERS FOR IMAGE RECOGNITION AT SCALE”

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Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, Neil Houlsby^{*,†}

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ABSTRACT

While the Transformer architecture has become the de-facto standard for natural language processing tasks, its applications to computer vision remain limited. In vision, attention is either applied in conjunction with convolutional networks, or used to replace certain components of convolutional networks while keeping their overall structure in place. We show that this reliance on CNNs is not necessary and a pure transformer applied directly to sequences of image patches can perform very well on image classification tasks. When pre-trained on large amounts of data and transferred to multiple mid-sized or small image recognition benchmarks (ImageNet, CIFAR-100, VTAB, etc.), Vision Transformer (ViT) attains excellent results compared to state-of-the-art convolutional networks while requiring substantially fewer computational resources to train.¹

**Q2: How can an architecture
originally built to read text
possibly understand an image?**

Vision Transformer (ViT): A new paradigm

The idea is almost poetic:

Just as a sentence can be split into words (tokens), an image can be split into patches — and each patch can be treated like a word.

Published as a conference paper at ICLR 2021

“AN IMAGE IS WORTH 16X16 WORDS: TRANSFORMERS FOR IMAGE RECOGNITION AT SCALE”

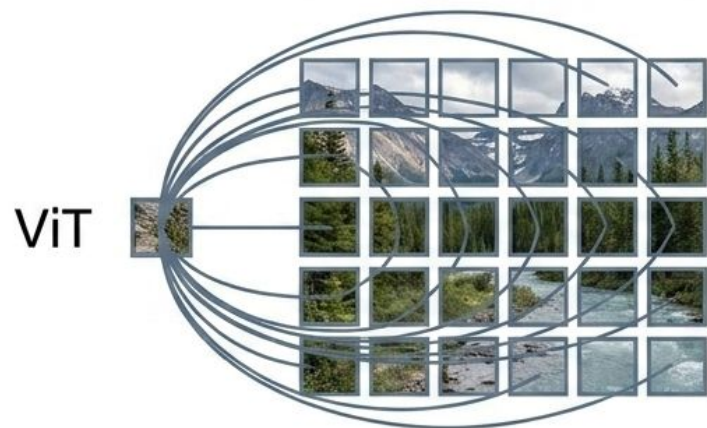
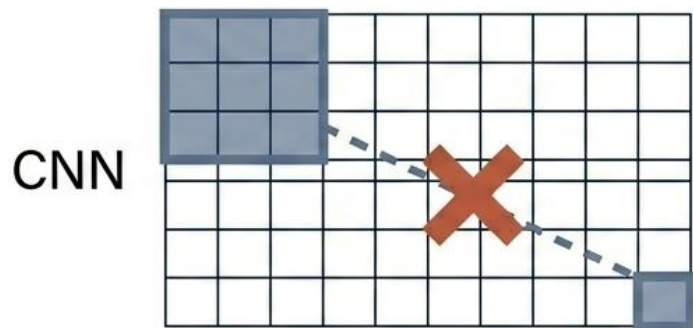
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Escaping Inductive Bias



- **Zero Locality Assumptions**

ViTs do not assume neighboring pixels are more important than distant ones.

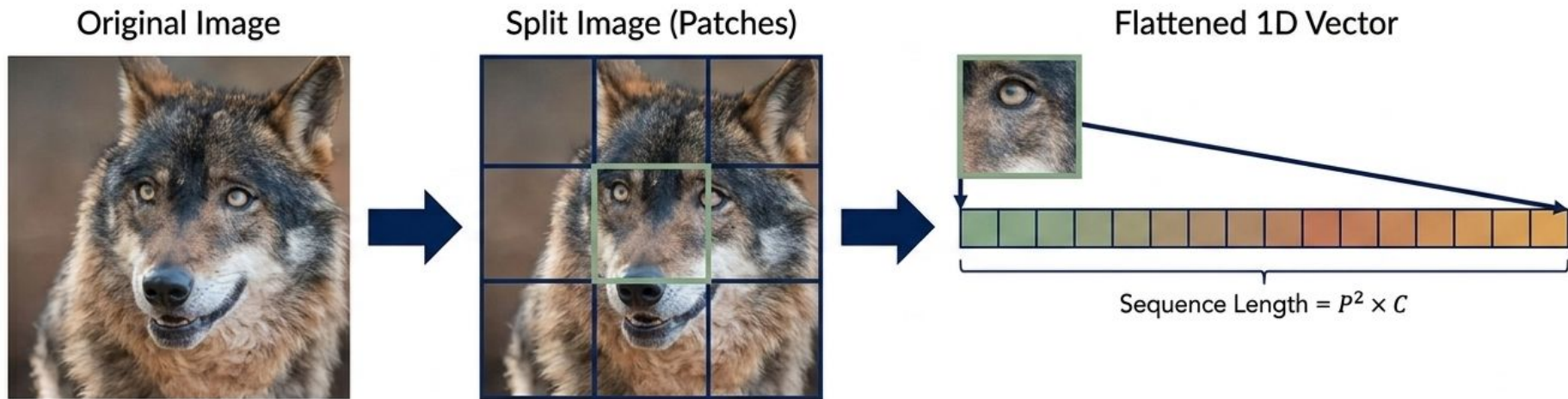
- **Immediate Global Context**

A patch on the top-left can attend directly to a patch on the bottom-right in the very first layer.

- **The Trade-Off**

ViTs remove the baked-in knowledge of the grid, trading spatial assumptions for complete architectural flexibility.

Step 1: Image Patching & Flattening



Patch Splitting

The 2D image is sliced into fixed-size, non-overlapping patches. A 48x48 image yields exactly 9 patches (16x16 pixels each).

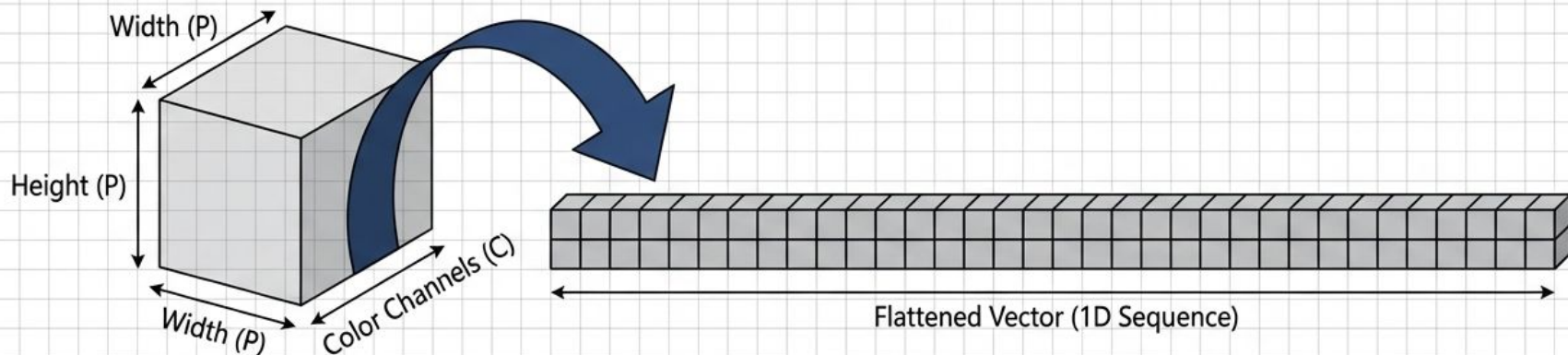
Dimensional Flattening

Each patch of size $P \times P \times C$ is mathematically flattened into a single 1D vector of length $P^2 \times C$.

The Purpose

Spatial dimensions are temporarily removed to create uniform, linear inputs that a Transformer can process.

Step 2: Patch Flattening



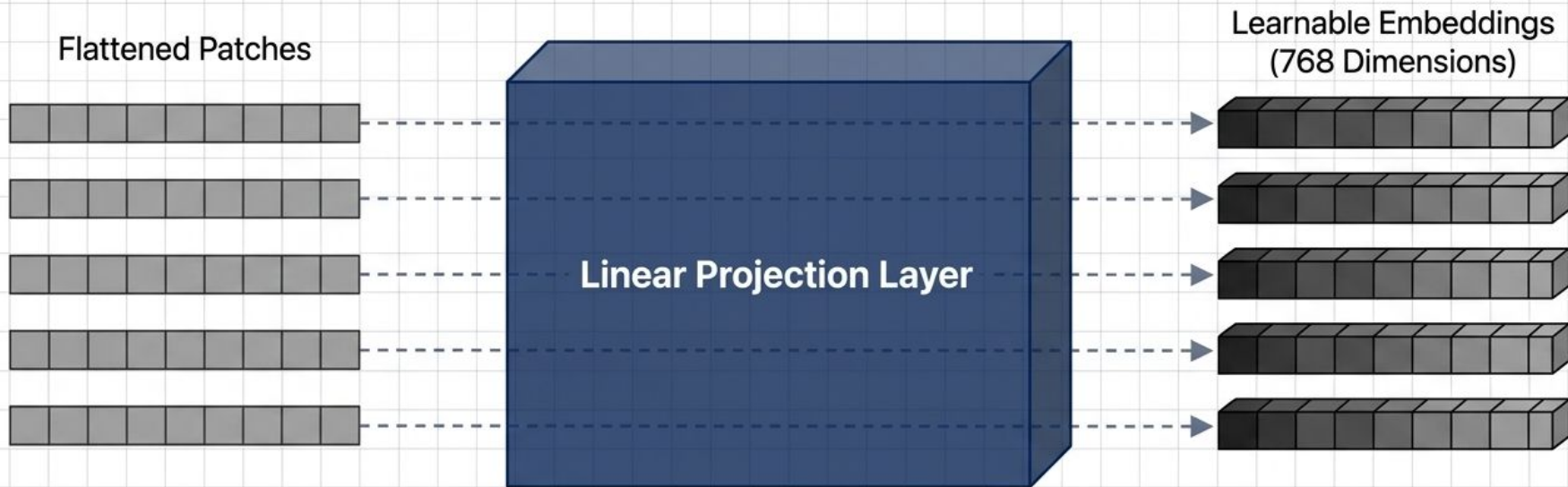
Removing Space

Spatial dimensions are temporarily stripped away. Each individual 3D patch of size $P \times P \times C$ is mechanically unrolled into a uniform 1D sequence of numbers.

Bridging the Gap

This mathematical flattening transforms varied spatial visual data into standardized vectors. The patch is no longer a square of pixels; it is now the exact raw format required by a language-style Transformer.

Step 3: Linear Projection (Embeddings)



Creating Tokens

Raw flattened vectors are mathematically mapped to a learnable, fixed-dimensional embedding (commonly 768 dimensions in base models).

High-Level Features

This projection enables the model to begin learning visual features immediately. The raw image patches have now fully transitioned into true 'tokens,' functionally identical to word embeddings in NLP.

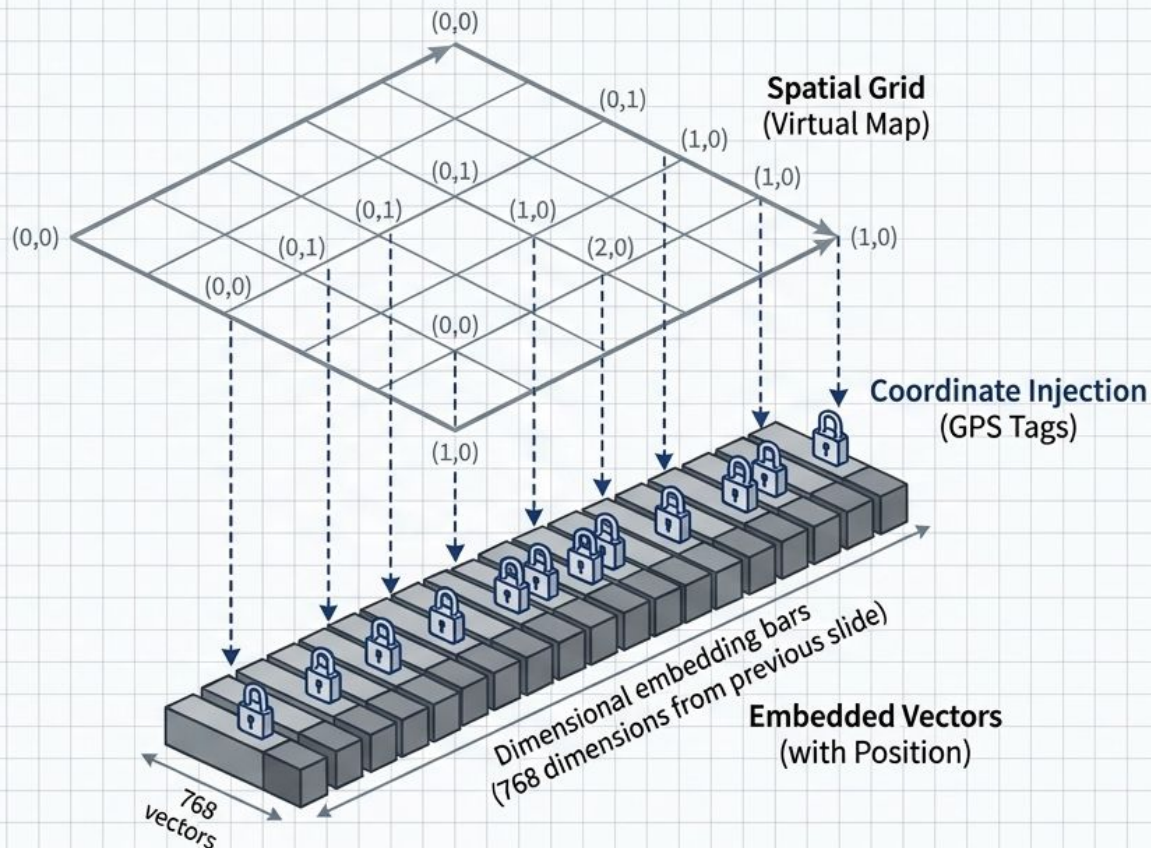
Step 4: Positional Embeddings

The Memory Problem

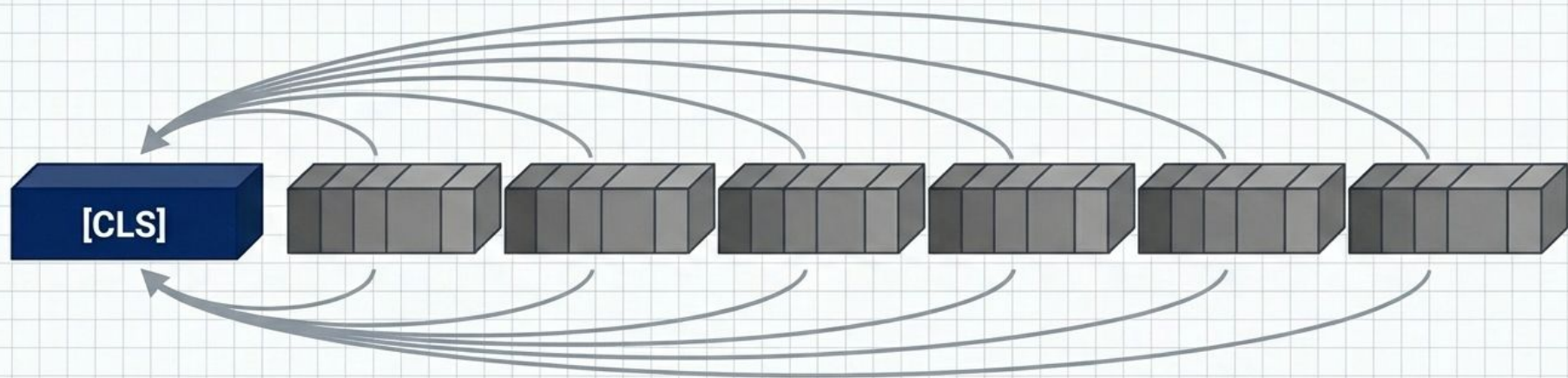
Because Transformers process **all tokens simultaneously** (they are **permutation invariant**), the network has **zero concept of 2D space**. It does not know where any patch originated.

Injecting Coordinates

To solve this, **learnable Positional Embeddings** acting as virtual “GPS tags” are injected directly into each vector. This allows the model to **permanently lock down** the original local and **global spatial layout** without relying on rigid CNN grids.



Step 5: The CLS Token



The Introduction of the Global Sponge

A final, specialized Classification Token ([CLS]) is prepended to the very start of the sequence.

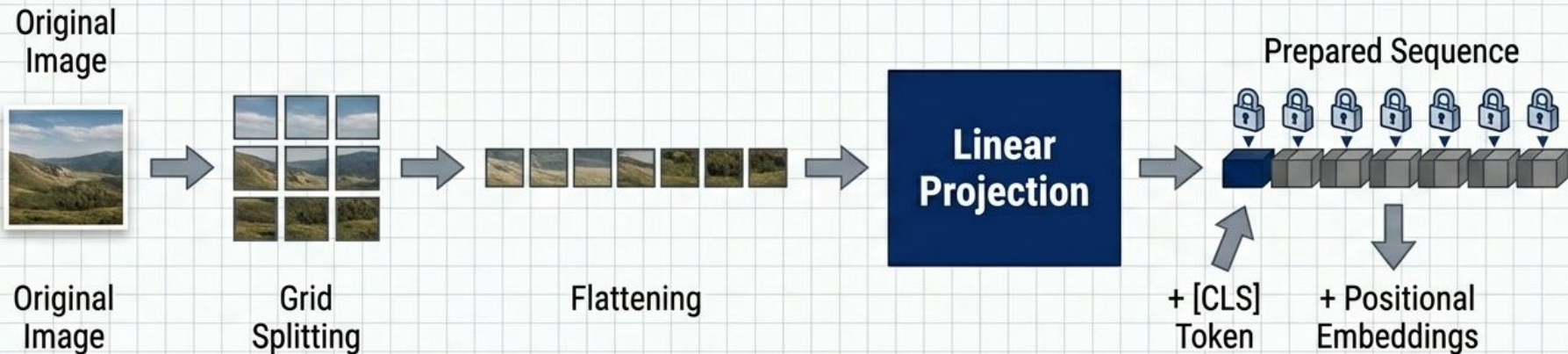
Information Aggregation

As this special learned token passes through the deep Transformer layers, its sole job is to act as an information sponge. It constantly aggregates global context from every other patch in the sequence.

The Final Output

When the processing is complete, the network ignores the image patches and uses only this deeply contextualized [CLS] token to serve as the singular, final classification output.

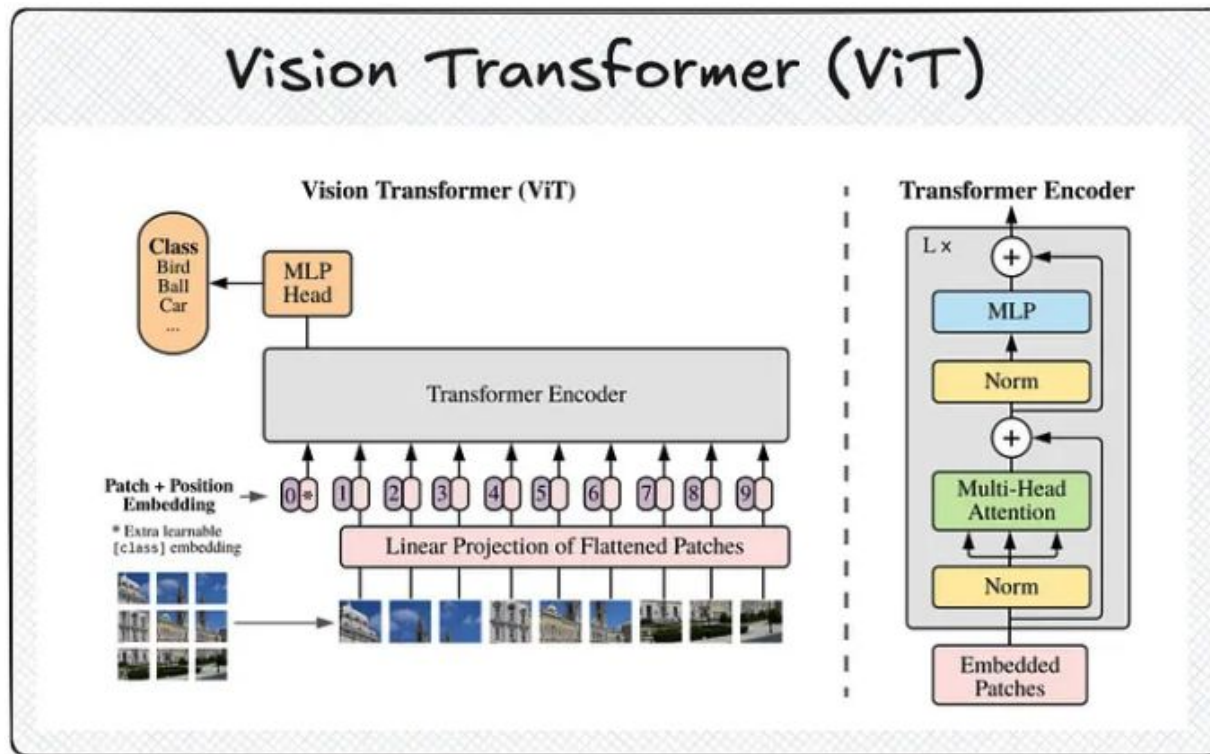
The Complete Input Pipeline



Synthesis: Zero Convolutions

A spatial 2D image has been systematically converted into a 1D sequence of context-rich tokens. Up to this point, prior knowledge (inductive bias) is kept at absolute zero, forcing the model to learn the spatial relationships entirely from the data. The sequence is now ready for the Transformer Encoder.

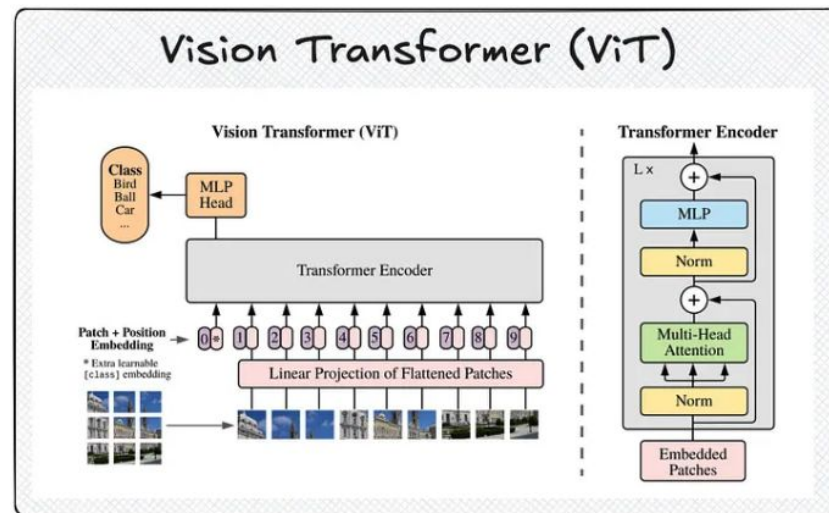
The Full ViT Architecture



The Full ViT Architecture

The overall structure of the vision transformer architecture consists of the following steps:

1. Split an image into patches (fixed sizes)
2. Flatten the image patches
3. Create lower-dimensional linear embeddings from these flattened image patches
4. Include positional embeddings
5. Feed the sequence as an input to a SOTA transformer encoder
6. Pre-train the ViT model with image labels, then fully supervise on a big dataset
7. Fine-tune the downstream dataset for image classification



Multi-Head Self-Attention: The Engine of Global Context



Dynamic Interaction

Unlike a CNN filter trapped in its 3×3 box, MHSA forces every single patch to mathematically query every other patch in the sequence simultaneously.

Relevance Mapping

It dynamically determines which parts of the image are most contextually relevant to each other, regardless of physical distance.

Multiple Heads

Multiple attention 'heads' run in parallel, tracking different relational aspects (e.g., color patterns vs. geometric edges) to yield a deep, unified interpretation.

**Q3: But without CNNs' built-in
spatial assumptions, don't
Transformers
require an impossibly massive
amount of data to learn?**

III:Data efficient Image Transformer

The Core Concept

DeiT introduces a novel architecture that maintains the powerful global context of the Vision Transformer (ViT) but completely eliminates the reliance on massive, proprietary external datasets.

The Secret Mechanism: A 'Teacher-Student' distillation strategy allows the data-hungry Transformer to rapidly inherit structural visual logic from a highly efficient CNN.

ImageNet Only

Training Data Requirement

Zero external or proprietary data required, completely bypassing the need for massive datasets like JFT-300M.

Single Node

Compute Power

Drastically lowers the hardware barrier to entry compared to standard ViT pre-training.

< 3 Days

Time to Train

Rapid iteration cycle allowing for faster downstream fine-tuning and deployment.

The Distillation Strategy: CNN as the “Teacher”

Step 1: The Teacher (Pretrained CNN)



CNN Prediction

A standard CNN, already optimized and rich with inductive biases, processes the image to generate a baseline prediction.

Step 2: The Student (Transformer)



Transformer Encoder

[DISTIL]
token

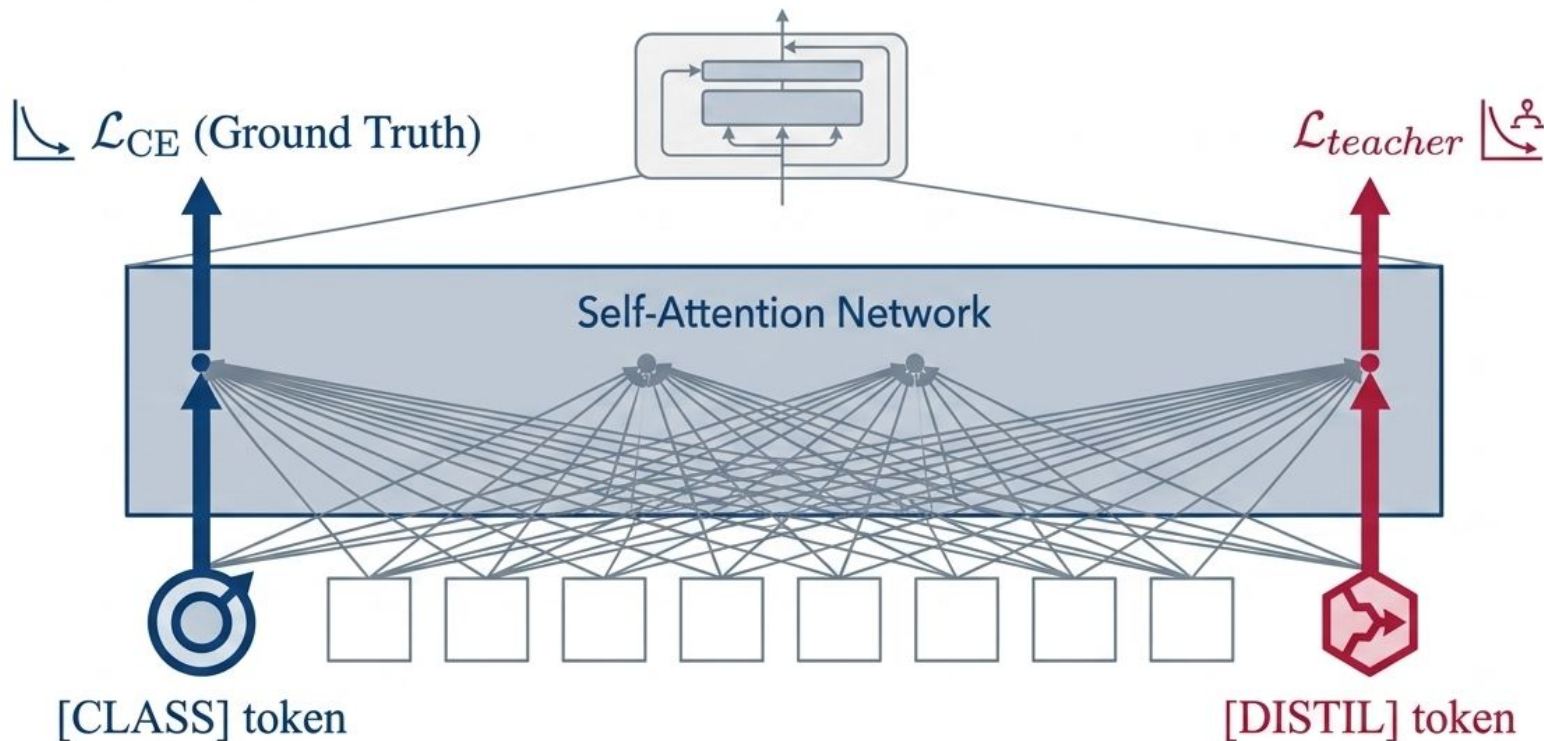
The ViT model attempts to process the exact same image patches from scratch.

Step 3: The Knowledge Transfer

Distillation
Loss

Instead of only learning from the raw ground-truth label, the Student actively analyzes and attempts to mimic the Teacher's outputs, drastically accelerating its own learning process.

The Distillation Token: A Dedicated Channel for Teacher Knowledge



Unlike standard distillation applied only at the final output, the distillation token participates actively in the network. It interacts with class and patch tokens through every self-attention layer, learning to explicitly reproduce the Teacher's hard labels via back-propagation.

Soft Distillation: Learning the Teacher's "Nuance"

The Concept

- **Objective:** Soft distillation evaluates the teacher's confidence across all possible options (the softmax probabilities), rather than just the final answer.
- **The 'Dark Knowledge':** It teaches the student that while an image is primarily a 'dog,' its secondary visual characteristics are mathematically closer to a 'cat' than a 'car,' transferring richer structural knowledge.

The Mathematical Mechanism

$$\mathcal{L}_{\text{global}} = (1 - \lambda) \mathcal{L}_{\text{CE}}(\psi(\mathbf{Z}_s), \mathbf{y}) + \lambda \tau^2 \text{KL}(\psi(\mathbf{Z}_s/\tau), \psi(\mathbf{Z}_t/\tau))$$

- **KL Divergence:** Minimizes the difference between the teacher's logits ($\mathbf{Z}_t$) and the student's logits ($\mathbf{Z}_s$).
- **Temperature (τ):** A smoothing parameter that softens the teacher's probability distribution, bringing secondary similarities to the surface.
- **Balancing Coefficient (λ):** Weights the priority of learning from Ground Truth vs. learning from the Teacher.

Hard-Label Distillation: Following Decisive Guidance

The Concept

- **Objective:** Instead of learning nuanced probabilities, the student treats the teacher's absolute top choice as the indisputable truth.
- **Data Augmentation Synergy:** Aggressive augmentations (cropping, skewing) can completely change an image's primary subject. The teacher's hard label dynamically updates to reflect these changes, and the student learns this dynamic behavior directly.
- **No Smoothing:** Unlike traditional distillation models, DeiT explicitly avoids smoothing the pseudo-labels provided by the teacher in this setup.

The Mathematical Mechanism

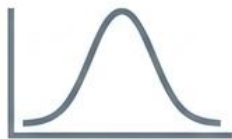
$$y_t = \operatorname{argmax}_c Z_t(c)$$

$$\mathcal{L}_{\text{global}}^{\text{hardDistill}} = \frac{1}{2} L_{\text{CE}}(\psi(Z_s), y) + \frac{1}{2} L_{\text{CE}}(\psi(Z_s), y_t)$$

- **Argmax (y_t):** Extracts the single highest probability prediction from the teacher.
- **Dual Cross-Entropy (L_{CE}):** The model is penalized equally for deviating from the true ground label (y) and the teacher's hard label (y_t).

Distillation Paradigms: Soft Probabilities vs. Hard Decisions

Soft Distillation



Objective

Minimizes Kullback-Leibler (KL) divergence between the softmax distributions of the Teacher and Student.

Key Variable

Utilizes a Temperature parameter (τ) to soften the probability distribution.

Hard-Label Distillation



Objective

$$y_t = \operatorname{argmax}_c Z_t(c)$$

Treats the Teacher's absolute highest-confidence prediction as the absolute ground truth.

Key Variable

No label smoothing. The teacher's label adapts dynamically based on specific data augmentations applied to the image.

DeiT authors opt for Hard-Label distillation, bypassing the need to mimic smoothed probability curves in favor of explicit decision boundaries.

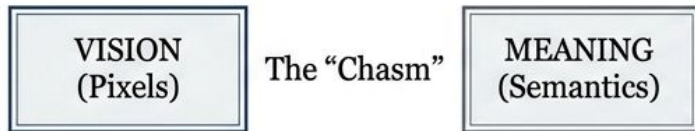
The Modality Chasm: Bridging Words and Pixels

I. The Legacy Problem: Modality Isolation

A. Computer Vision



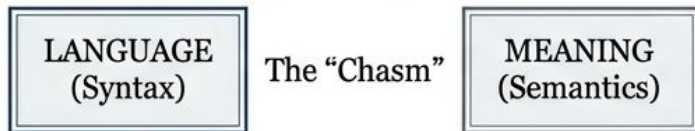
Identified pixel patterns but had no semantic understanding of real-world meaning.



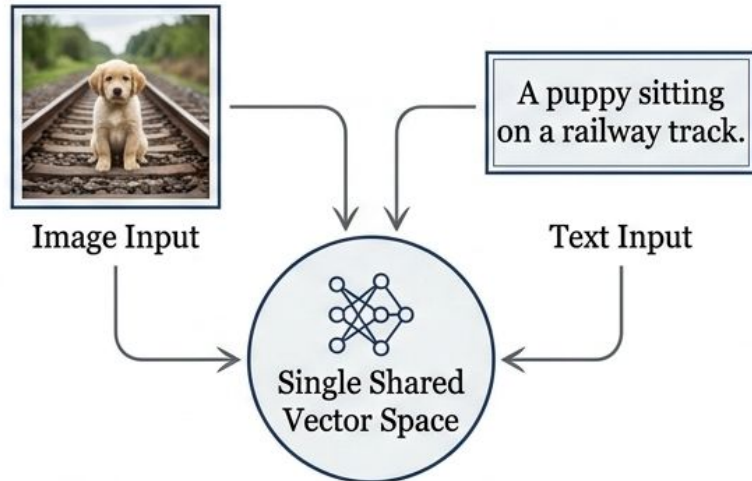
B. Natural Language Processing (NLP)



Processed text syntax flawlessly but was entirely blind to visual reality.



II. The Solution: CLIP (Contrastive Language-Image Pre-training)

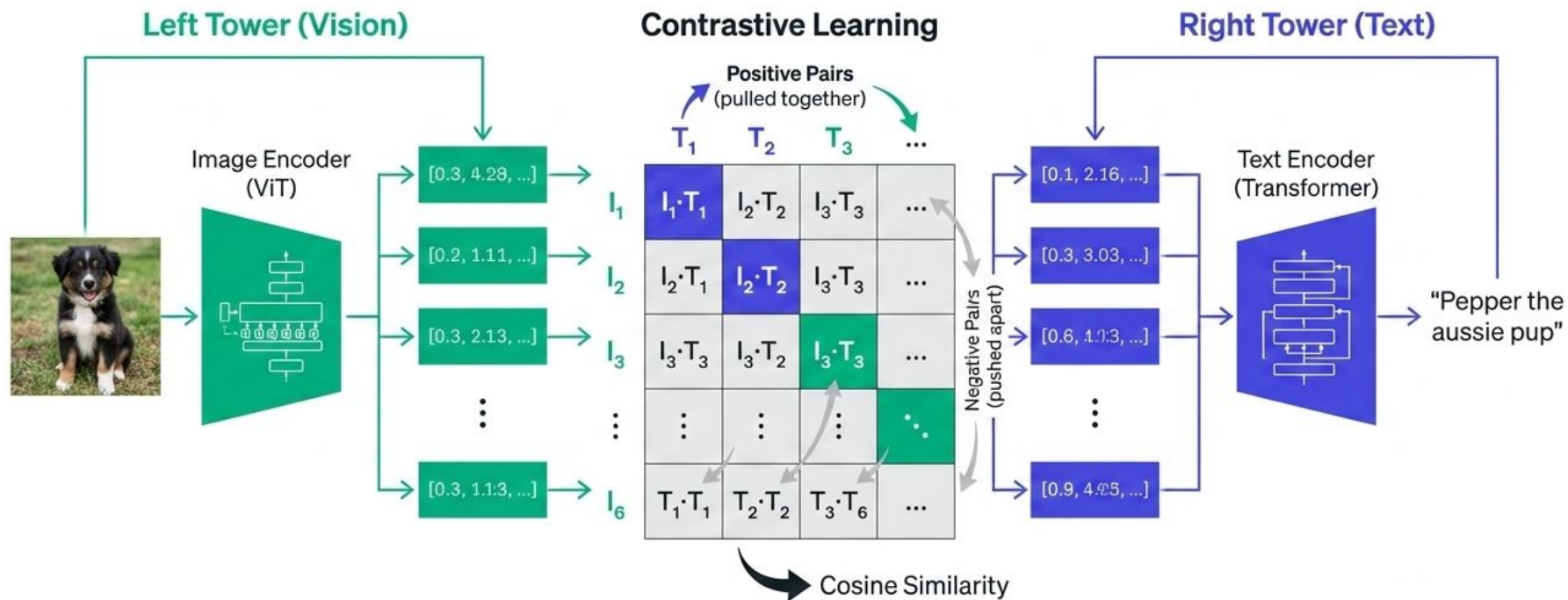


Machines no longer just classify pixels; they perceive visual inputs semantically, linking them directly to human language and broad world knowledge.

IV: Cross Model Embeddings

Q4: If both text and images are now just ‘tokens’, can we bridge the gap betws how a machine reads and how it sees?

A4: Yes. CLIP maps language and vision into a shared mathematical space.



Introduction to CLIP

- CLIP, which stands for Contrastive Language-Image Pre-training, is a revolutionary AI model from OpenAI that has fundamentally changed how machines perceive our world.
- It bridges the gap between images and words, creating a powerful new way for AI to learn and understand. This post will explain what CLIP is, how it works, why its abilities are such a game-changer, and the incredible applications it has unlocked.



“A puppy sitting on a railway track.”

The “Two-Part Brain” Architecture

Left Brain (Image Encoder)



Vision Transformer
(ViT) or ResNet

Compresses visual data
into a mathematical
‘fingerprint’.

$[0.4, -0.1, 0.8, 1.2\dots]$

Right Brain (Text Encoder)

A photo of a dog

Standard
Transformer

Compresses semantic
language into a
mathematical ‘fingerprint’.

$[0.4, -0.1, 0.8, 1.2\dots]$

=

The system utilizes entirely separate encoders for vision and language. The core objective is for semantically identical concepts (a picture of a dog, and the word ‘dog’) to produce mathematically identical dense vectors.

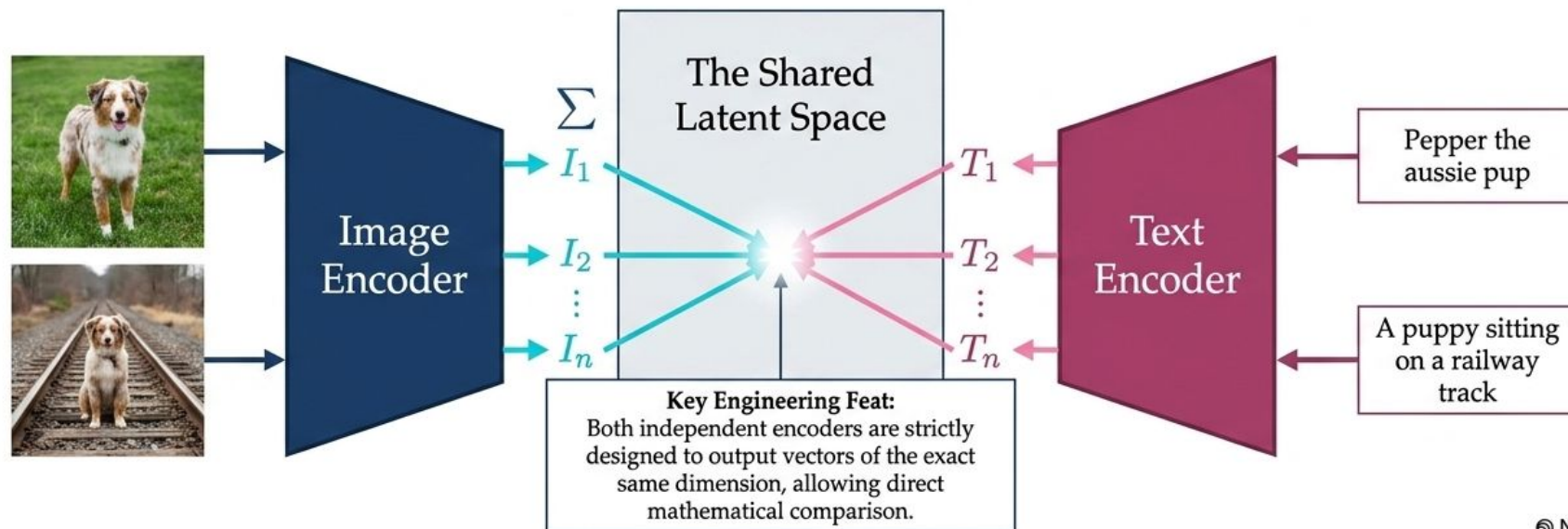
The CLIP Architecture: A Dual-Encoder System

The Image Encoder

- **Role:** The Computer Vision Expert.
- **Architecture:** Typically a Vision Transformer (ViT) or ResNet.
- **Function:** Ingests raw images and compresses them down into a dense mathematical vector (a numerical “embedding” fingerprint).

The Text Encoder

- **Role:** The Language Expert.
- **Architecture:** Standard Text Transformer.
- **Function:** Ingests text snippets and converts them into a semantic numerical embedding.



The Training Mechanism: Contrastive Learning

The Matching Game

In a massive batch of N images and N texts, the model calculates the Cosine Similarity for all possible pairings simultaneously to map corresponding items to the exact same geometric location in the latent space.

Image Encoders

Text Encoders

	T_1	T_2	...	T_N
T_1	$I_1 \cdot T_1$		...	
T_2		$I_2 \cdot T_2$	...	
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
I_N			...	$I_N \cdot T_N$

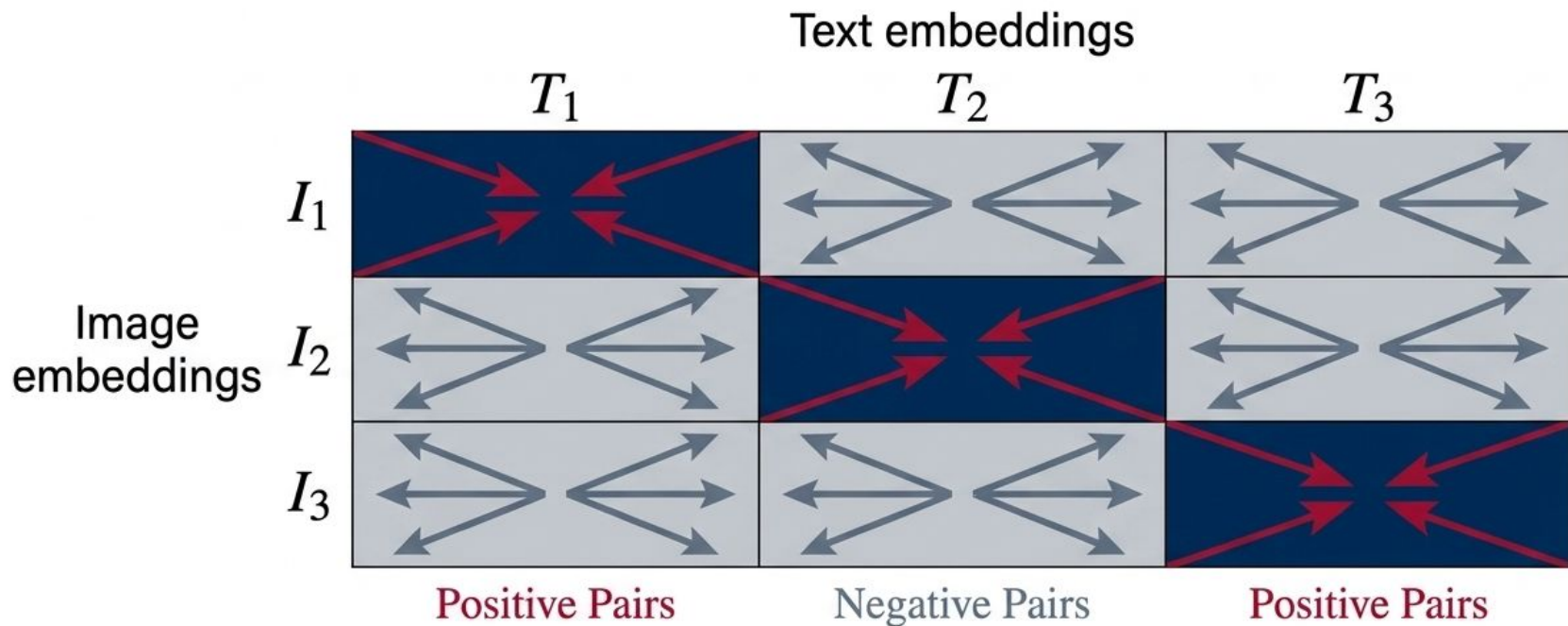
Push (Positive Pairs)

The model is mathematically rewarded for maximizing the similarity of true image-text pairs, pushing their vectors closer together.

Pull (Negative Pairs)

The model is penalized via InfoNCE Loss for associating unmatched images and text, pulling their vectors further apart.

The Contrastive Learning Engine



Powered by InfoNCE loss and Cosine Similarity, the model plays a massive matching game. It is rewarded for “pushing” the vectors of true image-text pairs together, and heavily penalized if it fails to ‘pull’ mismatched pairs apart.

THE OUTCOME: UNIFIED CROSS-MODAL REASONING

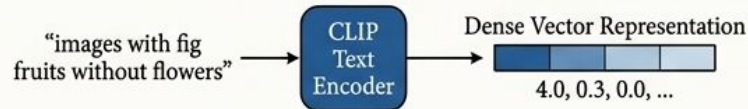
A Step-by-Step Lecture on the CLIP Search Process

1. TEXT QUERY

USER INPUT:
“images with fig fruits without flowers”

- A user inputs a highly specific semantic search query, including complex concepts and negations.
- This initiates the reasoning process.

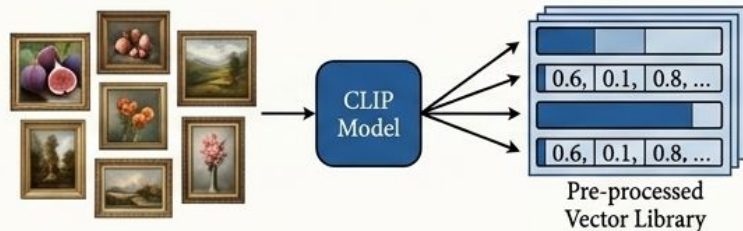
2. TEXT VECTORIZATION



- The CLIP Text Encoder compresses the exact semantic meaning into a dense mathematical vector.
- This is a unique “embedding” for the query.

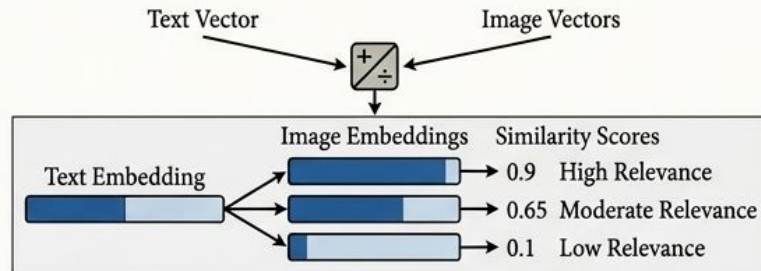


3. IMAGE BANK VECTORIZATION



- A massive collection of database images has already been pre-processed.
- This creates a comprehensive library of corresponding vector embeddings.

4. RELEVANCE CALCULATION



- A simple mathematical calculation instantly retrieves and ranks images based on vector similarity.
- Results are ordered by relevance (e.g., 0.9 to 0.1).

**Q5: What is the ultimate
“superpower” unlocked by
this language-image fusion?**

A5: Zero-Shot Learning (Generalization without retraining).



Zebra

1	Stripes	✓	a photo of a zebra
1	Hooves	✓	
1	Mane	✓	
1	Fur	✓	
0	Wings	✗	

A traditional CNN needs a curated dataset of 1,000 zebra photos to learn.

CLIP sees an image and correctly identifies "a photo of a zebra" purely based on concepts learned from internet text, having never been explicitly trained on a zebra dataset.

Image Generation

Guiding DALL-E/Stable Diffusion text-to-image synthesis.

Semantic Search

Searching for context (e.g., "a black cat sleeping on a red sofa") without relying on metadata tags.

Robotics

Cross-modal understanding for physical commands like "pick up the red ball".

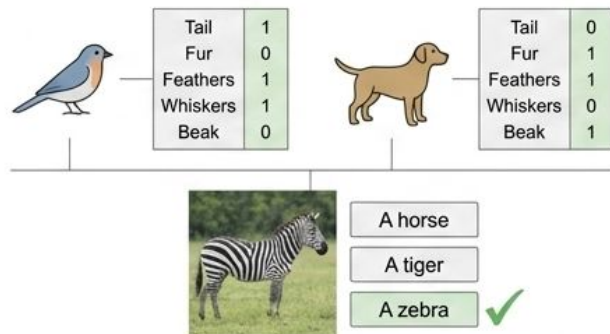
The CLIP Superpower: Zero-Shot Prediction

Legacy AI (Supervised Learning)

- **Requirement:** To classify a “Zebra,” the model required thousands of manually curated, explicitly labeled images in a specific “Zebra Dataset.”
- **Limitation:** If asked to identify an “Astronaut,” it would completely fail if that specific category was not in its pre-defined training taxonomy.



CLIP (Zero-Shot Learning)



- **Capability:** CLIP can be fed an image of a zebra and accurately identify it on the first try from multiple choice text prompts.
- **Mechanism:** Because it learned broad world knowledge by reading millions of generic internet image-caption pairs, it inherently already knows what a zebra is without explicit curated training.

Unlocking a New Generation of AI Applications

Guiding Image Generation

- CLIP serves as the internal 'critic' behind generative tools like DALL-E 2 and Stable Diffusion.
- As the generator creates an image, CLIP continuously calculates how well the pixels match the user's text prompt, actively steering the output toward accuracy.

Contextual Content Moderation

- Automatically flags harmful or inappropriate content by understanding the abstract concept and context of an image.
- Bypasses the limitations of relying on rigid, pre-defined visual categories that easily miss nuanced context.

Smarter Semantic Search

- Moving beyond basic keyword metadata, users can search for highly complex visual concepts.
- Example: A search for 'a black cat sleeping on a red sofa' retrieves visually accurate results based purely on mathematical semantic understanding.

Spatial Robotics

- Enables physical robots to successfully execute semantic human commands.
- Example: Instructing a robot to 'Pick up the red ball from the table' works by unifying raw visual sensor data with natural language processing instructions.

System Limitations: Where CLIP Falls Short

Fine-Grained Distinction

Excels at broad categorization (“a car”) but fails completely at highly specific domain knowledge.
Example: Cannot reliably distinguish a 2021 Honda Civic from a 2022 model visually.

Abstract Logic & Nonsense

Fails to process highly abstract, logical, or nonsensical prompts.
Reasoning: If a conceptual prompt lacks a clear, heavily repeated visual correlate in its internet-scraped training data, the vector embedding breaks down.

Mathematical & Spatial Reasoning

Struggles fundamentally with counting and evaluating exact spatial relationships.
Example: Understands the semantic concept of an “apple” but frequently fails to render or identify “a photo of exactly three apples.”

Inherited Bias

Actively inherits and amplifies human biases regarding race, gender, and culture.
Reasoning: Foundational ‘world knowledge’ is scraped directly from unfiltered internet data, permanently encoding societal prejudices into the mathematical latent space.

Cross-Modal Embeddings at Industry Scale

Google - Multisearch



ViT + T5
Multimodal Fusion

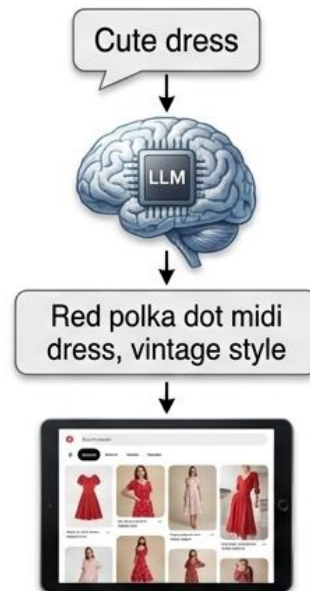
Unified Representation
& Retrieval

Meta - ImageBind



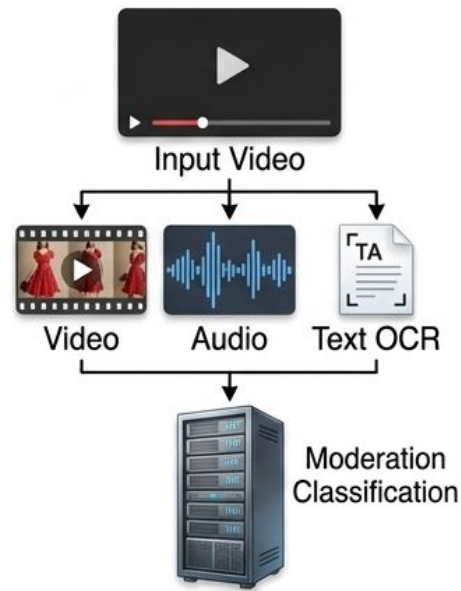
Joint Embedding Space

Pinterest - OmniSearchSage



Semantic Enrichment
& Discovery

TikTok - Content Processing



Safety &
Recommendation Signal

The multimodal paradigm has become the foundational infrastructure for perception across the world's largest consumer platforms.

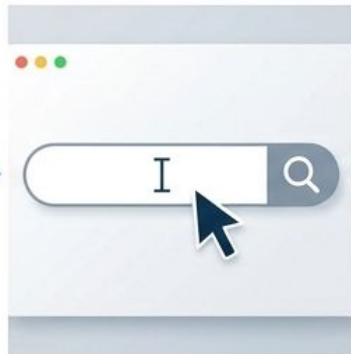
Operationalizing the Vector Space

Step 1: Offline Embedding



Pre-computing vectors for millions of database images.

Step 2: Online Query



Converting a user's text prompt into a vector in real-time.

Step 3: ANN Search



Approximate Nearest Neighbors (HNSW/IVF-PQ) algorithms locate closest matches in milliseconds.

Step 4: Re-Ranking



A Heavy Ranker (Cross-Encoder) deeply analyzes the top 50 matches to return the absolute best result.

Finding semantic matches among billions of images requires complex **approximate search algorithms** to avoid calculating every single possible pair.

Foundational Applications in the Wild

Generative Guidance



The steering brain behind DALL-E 2 and Stable Diffusion, scoring image-in-progress against the text prompt.

Semantic Search

Querying exact visual concepts without relying on metadata tags.



Content Moderation



Flagging harmful or inappropriate concepts natively, bypassing rigid pre-defined keyword lists.

CLIP is not just an endpoint classifier; it is a **foundational utility** that grants “**sight**” to other AI systems.

Robotics

Grounding human language (“pick up the red ball”) directly into the robot’s physical camera reality.



Limitations and Boundary Conditions

Fine-Grained Details



Struggles to distinguish micro-variations, such as the difference between a 2021 and 2022 car model.

Counting



Fails at exact numeric representation. Understands 'apples' but structurally fails at 'exactly three apples'.

Inherited Biases



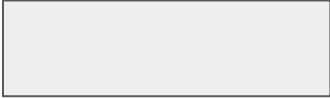
Trained on unfiltered internet data, inherently adopting and amplifying harmful human biases.

Abstract Concepts

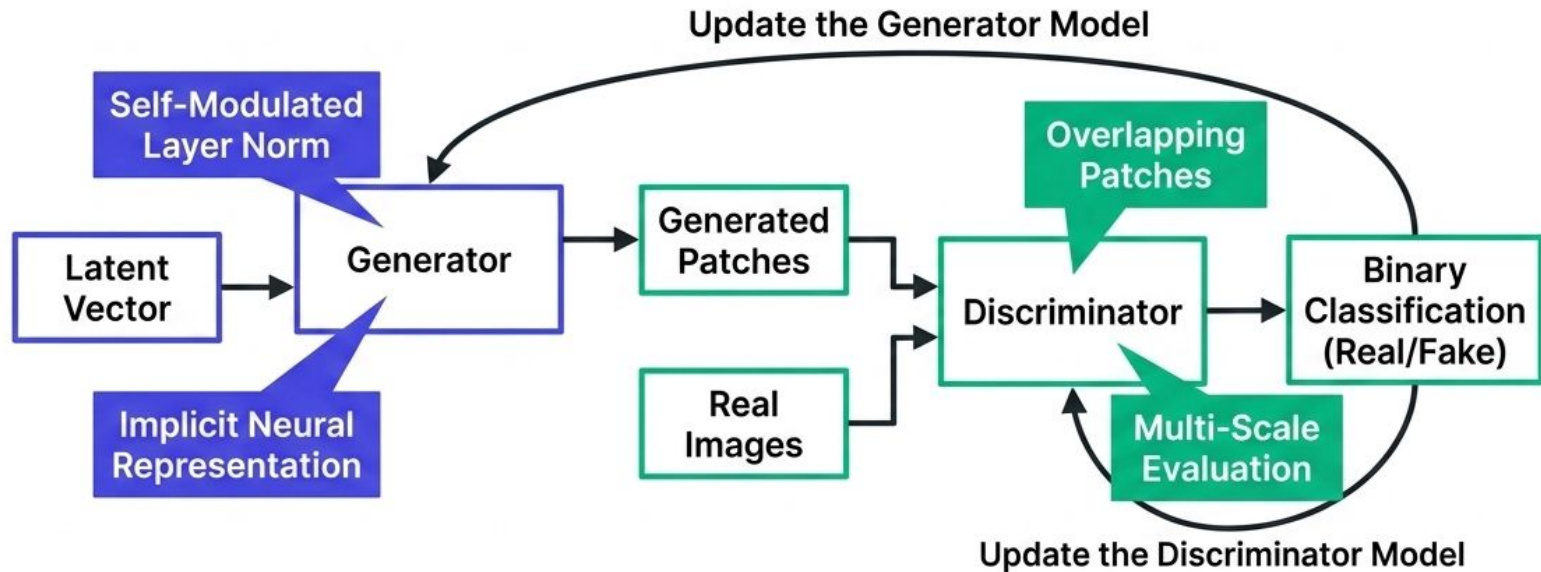


Lacks structural reasoning, failing on nonsensical or highly complex logical prompts.

While CLIP bridges the gap between vision and language, it remains an associative engine, not a deductive reasoning system.

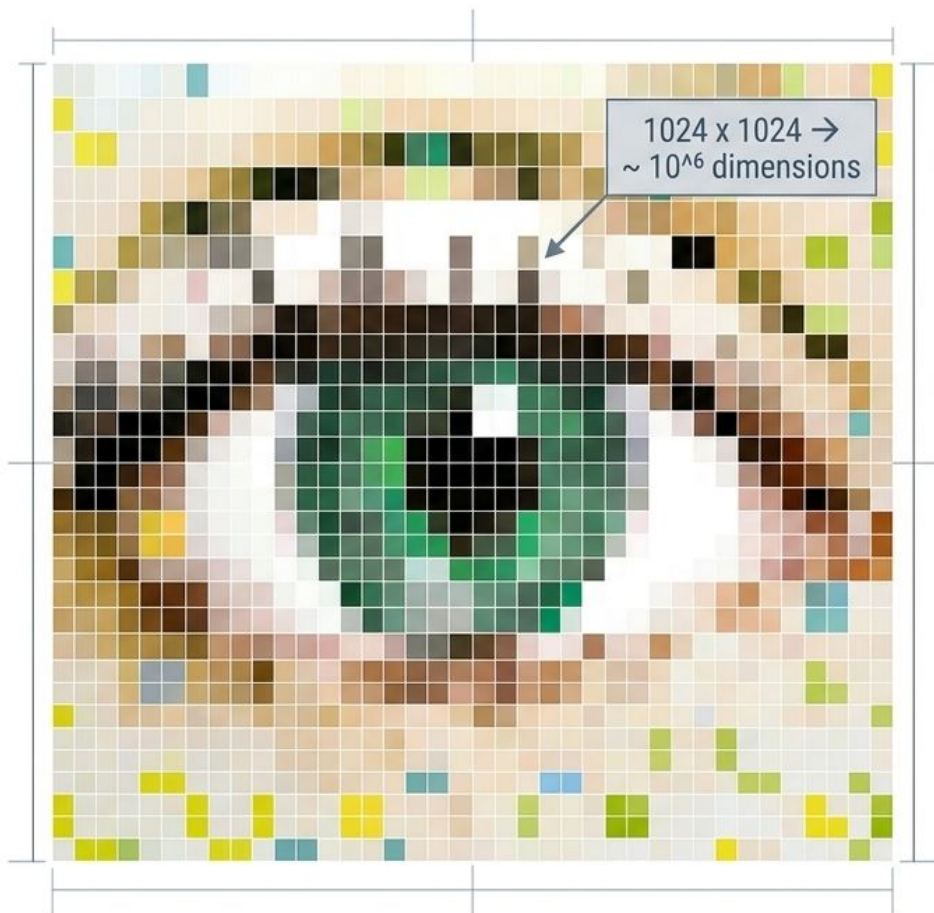
Q6: Analyzing reality is one thing—but can Vision Transformers be used to generate entirely new  realities?

A6: Yes, through ViTGANs, though they require specific architectural stabilizing.



Early ViTs failed as GANs because self-attention caused extreme training instability. By applying token-based gradient penalties and overlapping patches, ViTGANs stabilize the architecture to rival top CNN generators.

Autoencoder



Autoencoders: Deep Learning for Dimensionality Reduction & Feature Extraction.

The Curse of Dimensionality

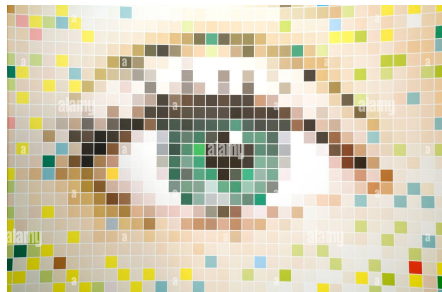
- Real-world data occupies a high-dimensional space.
- Its underlying useful information structure is much smaller.

Can we compress the data
without losing important signals?



Challenge of high-dimensionality

- Real world data is high-dimensional (image, audio, sensor data etc.).
- Useful information is much smaller.



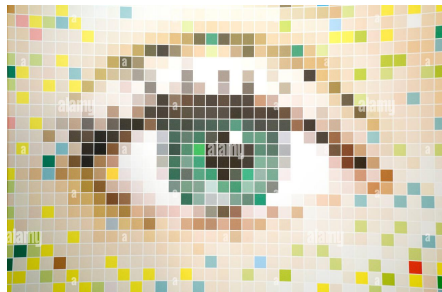
→ $\approx 10^6$ dimensions

1024x1024

4

Challenge of high-dimensionality

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→ $\approx 10^6$ dimensions

1024x102

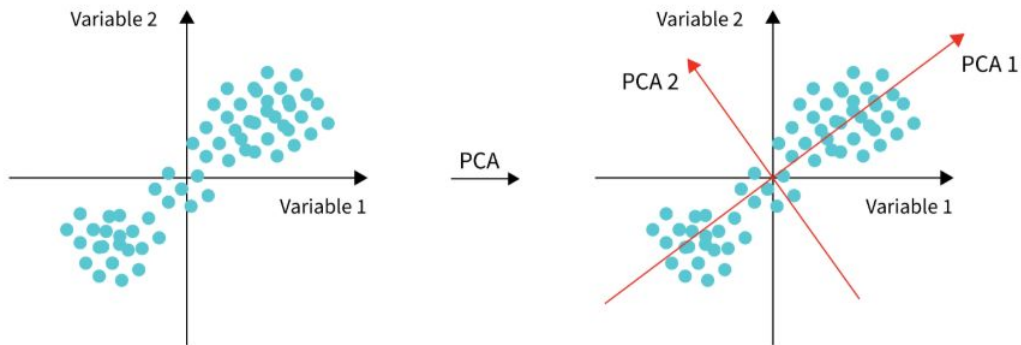
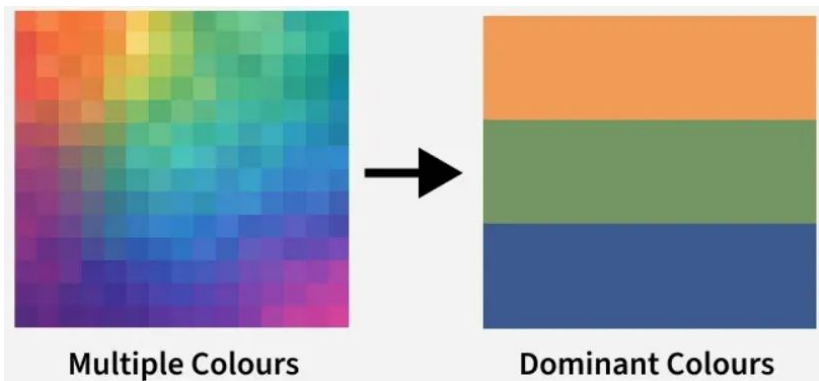
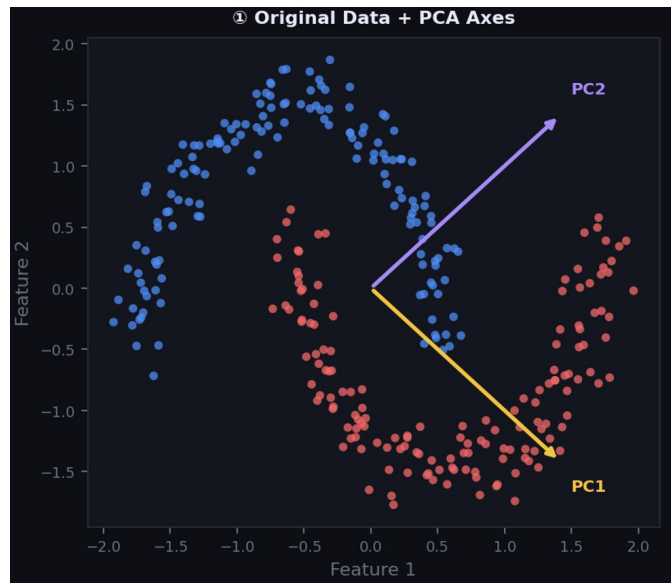
4



Can we compress the data without losing important signals?

But we had PCA?

- PCA is linear → Can't capture **complex non-linear** patterns.
- Real world data is often non-linear.



Autoencoders

- What if a neural network could learn how to compress data automatically – and then reconstruct it?

Autoencoders

- Supervised → Unsupervised
- Autoencoders solves:
 - Dimensionality reduction for non-linear data
 - Noise removal
 - Feature learning
 - Data compression
 - Anomaly detection

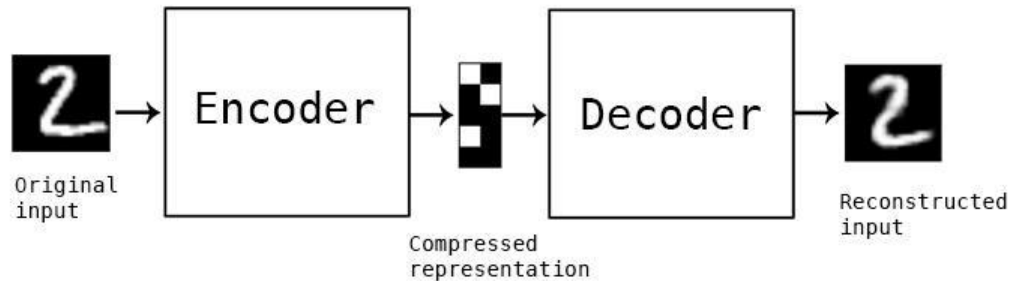
Autoencoders

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Autoencoders don't just reduce data, they learn what matters.

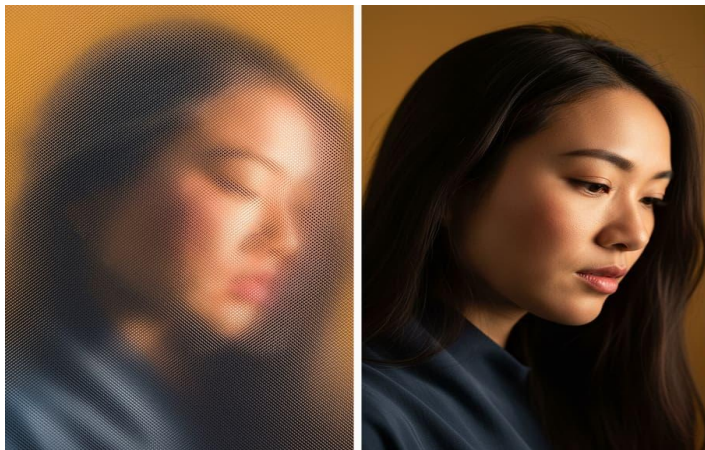
Autoencoders

- Supervised → Unsupervised
- Autoencoders transform unsupervised task to supervised task.
- Use inputs as the ground truth labels.



Autoencoders

- If you have to reconstruct this blurry image, what will you focus on?



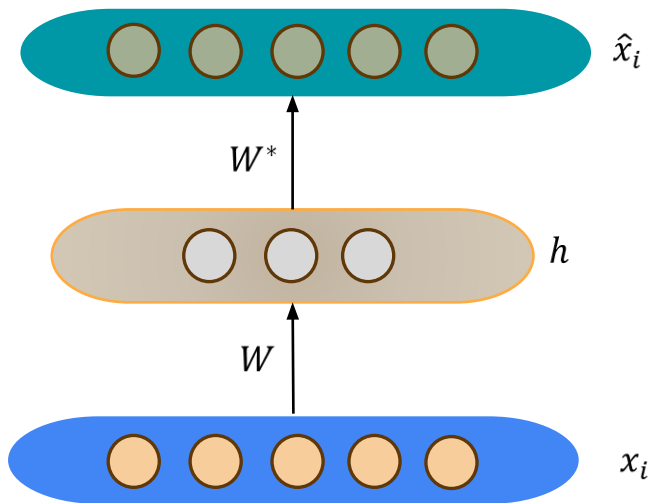
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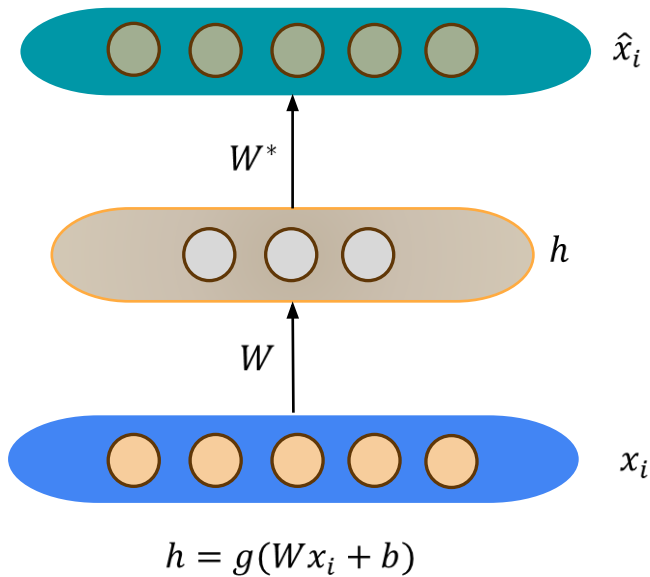
- Latent space
- Feature extraction
- Learning representation

Autoencoders



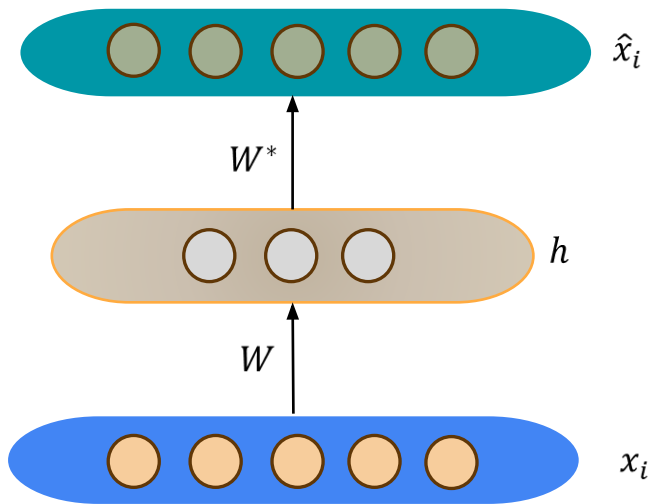
- A special type of feedforward neural network.

Autoencoders



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- **Encodes** its input x_i to a hidden representation h .

Autoencoders

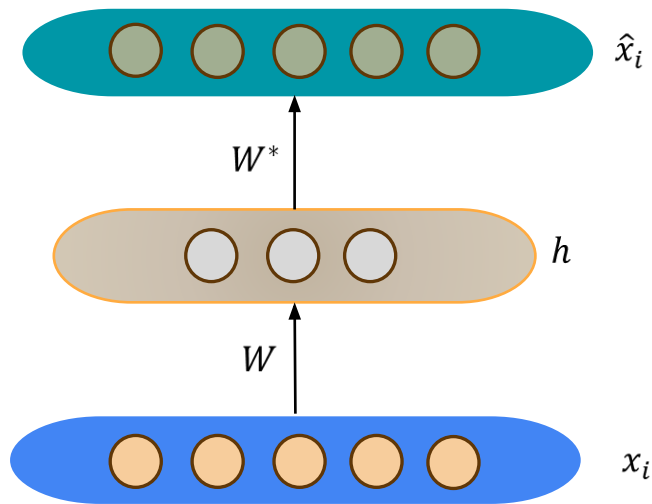


$$h = g(Wx_i + b)$$

$$\hat{x}_i = f(W^*h + c)$$

- A special type of feedforward neural network.
- **Encodes** its input x_i to a hidden representation h .
- **Decodes** the hidden representation h into the input.
- **Objective:** Make $\hat{x}_i$ closer to x_i .

Autoencoders

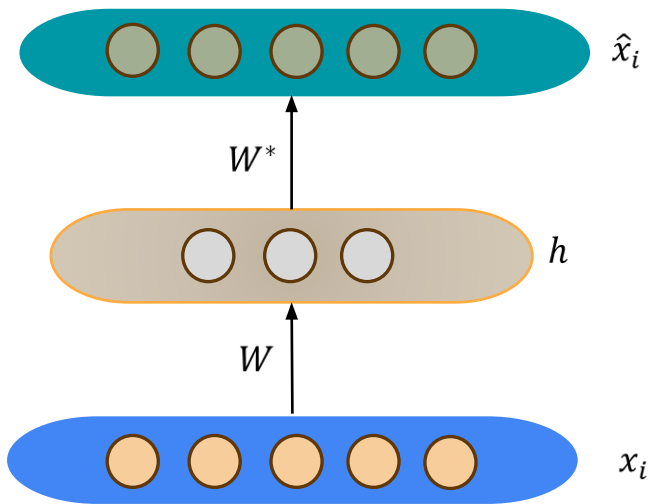


$$h = g(Wx_i + b)$$

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- Let $\dim(h) < \dim(x_i)$.
- If the NN is able to reconstruct $\hat{x}_i$ perfectly, what does it indicate about $\mathbf{h}$?

Autoencoders

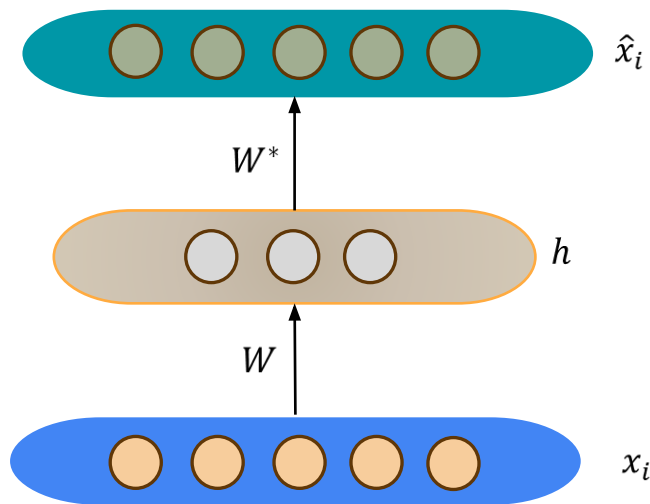


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- OR h represents all important characteristics of x_i .

Autoencoders



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An autoencoder where $\dim(h) < \dim(x_i)$ is called an **under complete autoencoder**.